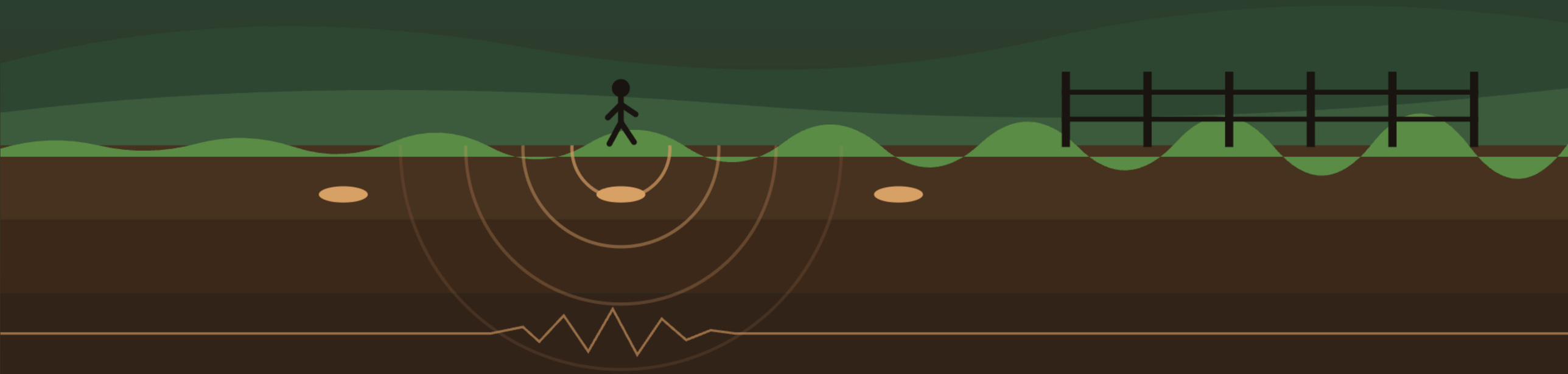
MPMC COURSE PROJECT || M2026

Multi-Sensor Intrusion Detection and Localization using the 8051 Microcontroller

Presented By:
Tishya Yadlapalli
Dhanvitha Sankuratri
Naseeb

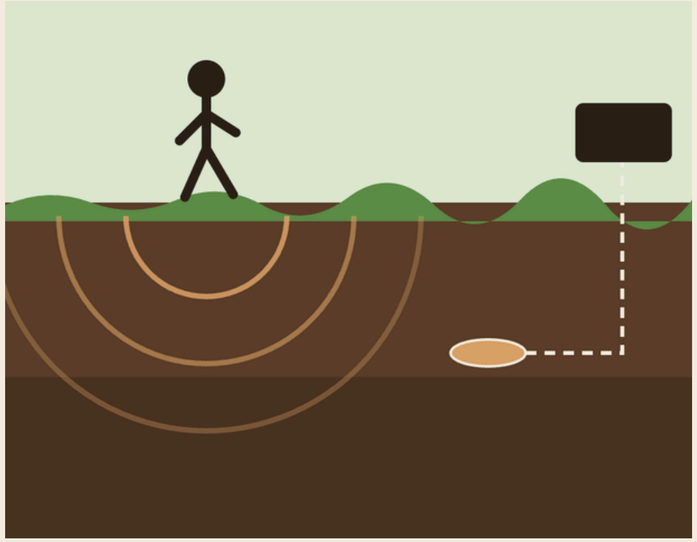


Introduction

Cameras need light and line of sight, and a single PIR gives false alarms without saying what moved or where. GroundWatch listens to ground vibration and cross-checks it with other sensors to tell what the intruder is and which part of the boundary they are at.

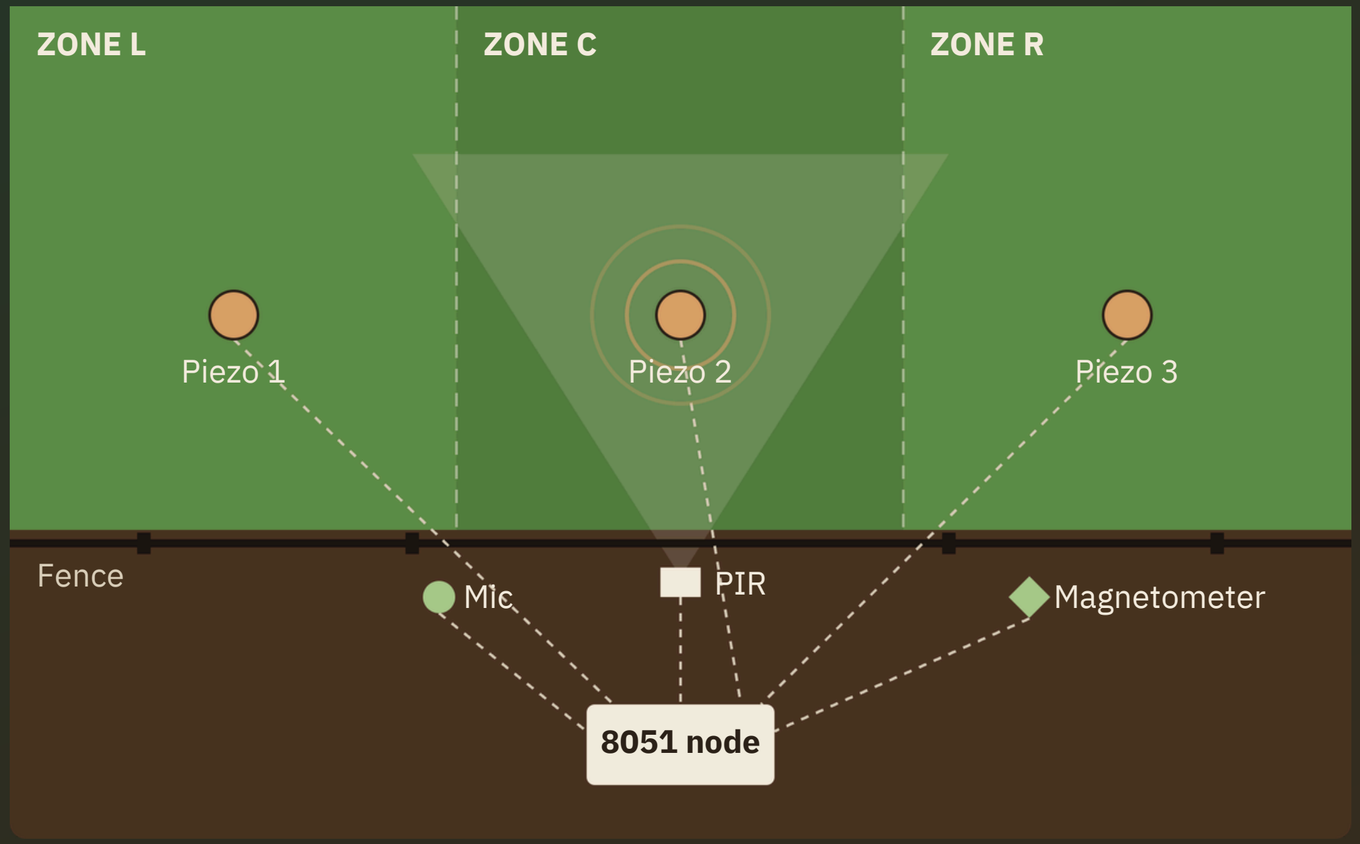
Objectives

- Detect and classify footsteps, vehicles, digging and animals near a boundary
- Locate the zone (left, centre or right) using three buried piezo sensors
- Combine vibration with PIR, magnetic and sound sensing to cut false alarms
- Run it all on an 8051 with a task scheduler, local alarm and PC logging



A footstep sends vibration through the soil to the buried piezo, which is wired to the 8051 node.

Sensor Layout



3× Piezo discs

Vibration strength and rhythm. The one that triggers first and strongest gives the zone.

PIR sensor

A warm body is present (person or animal).

Magnetometer

Change in magnetic field means a large metal object, usually a vehicle.

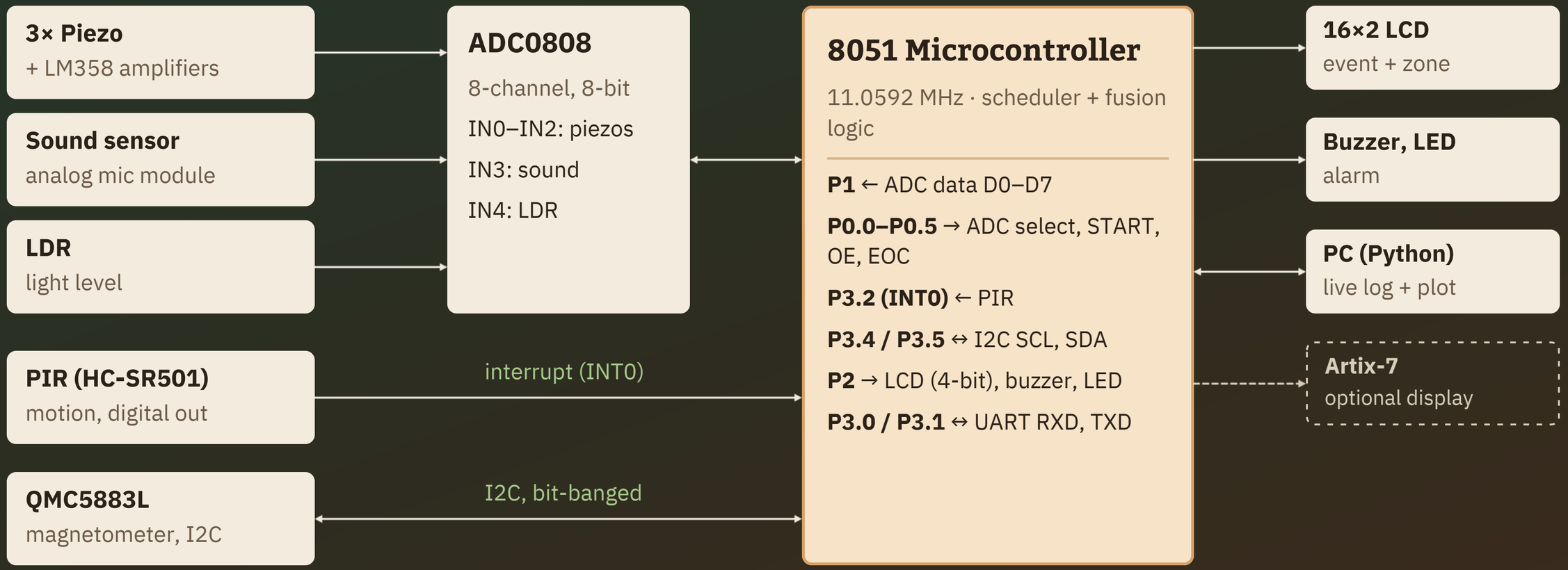
Sound sensor

Sharp impact noise from shovels and tools.

LDR

Day or night. Sensitivity is raised at night.

System Architecture



Planned pin map. Port 0 needs external pull-up resistors.

Firmware: Task Scheduler

Task	Runs every	What it does
Sample piezos	1 ms (in ISR)	Read IN0–IN2, update peak, peak count, active time, first-hit time
Sound + light	10 ms	Read IN3 and IN4, track sound bursts and day/night
PIR	on interrupt	INT0 sets a motion flag
Magnetometer	100 ms	Read X, Y, Z over I2C and compare with baseline
Classify	2 s	Fuse all flags, decide the event and zone
Display + alarm	200 ms	Update LCD, buzzer and LED
UART log	each window	Send event, zone and features to the PC

How it runs

Timer 0 gives a 1 ms tick. Each task has a period and a countdown, and the main loop runs whichever task is due. Piezo sampling sits inside the ISR, so a slow I2C read can never make it miss samples.

Memory limit

A basic 8051 has only 128 bytes of internal RAM, so raw samples are never stored. Each channel keeps running values (peak, count, active time, first-hit time) in a few bytes.

Working: Sensor Fusion

A piezo crosses its threshold → Zone = first and strongest hit → Check PIR, magnetometer, sound → Match the fusion table → **Alarm, LCD, UART log**

Event	Piezo pattern	PIR	Magnetometer	Sound	Action
Person	regular medium peaks	Yes	no change	low	Alarm + zone
Vehicle	long low rumble	any	large change	medium	Alarm + zone
Digging	strong irregular hits	Yes	no change	high bursts	Alarm + zone
Animal	weak, irregular	Yes	no change	low	Log only
Wind, rain	weak, random	No	no change	any	Ignore

Night mode: when the LDR reads dark, piezo thresholds drop higher sensitivity. Thresholds come from data we record and plot on the PC.

Components and Future Scope

Component	Use
8051 development board	Main controller
3× Piezo disc, 2× LM358	Vibration sensing + amplification
ADC0808	8-channel analog to digital
HC-SR501 PIR	Body detection
QMC5883L	Vehicle (metal) detection
Sound sensor module	Digging noise
LDR + resistor	Day / night mode
16×2 LCD, buzzer, LED	Local alert
USB-UART module	PC logging

Challenges:

- Environmental Noise
- Limited 8051 RAM
- Real-time processing
- Multi-sensor synchronization
- False positives



8051

40-pin chip with 4 I/O ports, 2 timers, a UART and 2 external interrupts. This project uses every one of them.

Future scope

Small neural network on the Artix-7 instead of the rule table, and LoRa alerts to a phone

THANK YOU